Ex.no: 1

Study of Various network Command used in Linux and Windows

AIM:

To study and observe various basic network commands used in Linux and Windows operating System for network configuration, monitoring

Basic Networking Commands (Windows/General)

1.) arp -a – Shows IP and MAC address of router
2.) hostname – Display the name of the computer
3.) ipconfig/all – Show detailed TCP/IP configuration
4) nbstat -a – Resolve NetBIOS names over TCP/IP
5) netstat – Displays active TCP/IP connection and statistic
6) netstat-r – Displays the routing table
7.) nslookup – perform DNS lookup for a domain/host
8.) pathping – combines ping and tracert; Shows route and packet/la per
9.) ping – Test connectivity between two nodes using ICMP
10.) route – Display/modifies the IP routing table

Linux Networking Commands

1.) ip address show – Displays IP assigned to interface
2.) ip address add – Assign an IP address to a interface
3) ip address del – Delete IP from interface
4.) ip link set eth0 up – Bring interface online
5) ip link set eth0 down – Bring interface offline
6) ip link set eth0 promisc on – Enable promiscuous mode on an interface
7) ip route add default via – add a default gateway
8) ip route add (via gateway) – adds a route to a network via gateway
9) ip route add (dev) – adds a route to a network via device
0) ip route delete – Deletes a specified route

11) ip route get - Shows the route taken to reach an IP

12) ipconfig - display network interfaces

13) mtr - combines ping + traceroute for live route

14) mtr -g - shows numeric ip address

15) mtr -b - Show both numeric ip and hostname

16) mtr -c - sets no. of pings to send

17) tcpdump -D - Lists available interface for packet capture

18) tcpdump -ieth0 - capture traffic on specific interface

19) tcpdump -ieth0 -c10 - capture a fixed number (10) of packets

20) tcpdump host - capture traffic to/from a specific host

21) tcpdump src host - Capture traffic coming from a specific host

22) tcpdump dst host - capture traffic going to a specific host

23) tcpdump net - capture traffic to/from a specific network

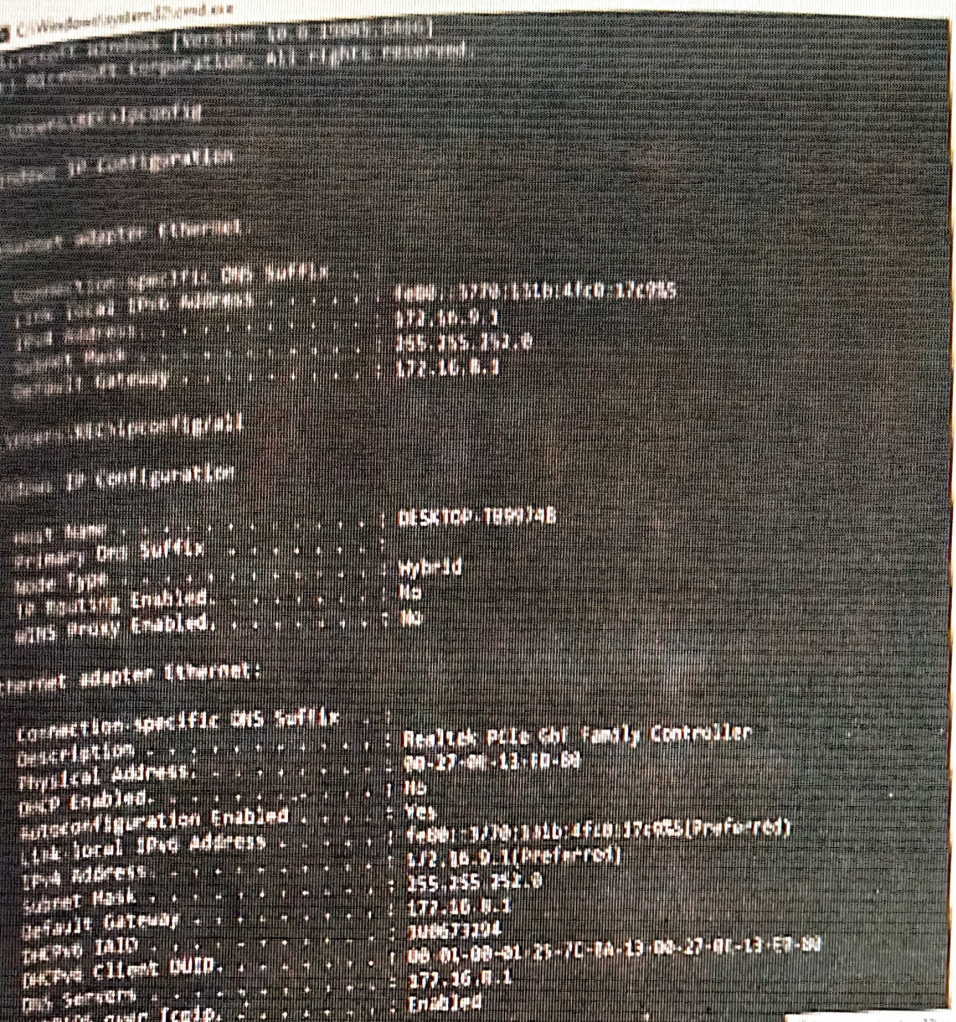
24) tcpdump port - capture traffic on a specific port

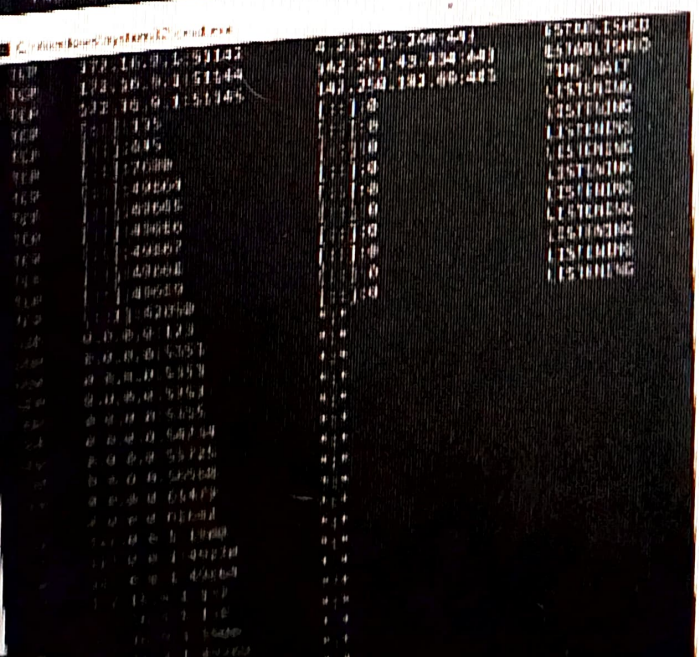
25) tcpdump port not - Excludes specific ports from capture

26) ping google.com - Check connectivity/reachability using ICMP

27) ping -c10 - send a fixed number (10) of ping request

Windows :


ipconfig
IP Configuration
Ethernet adapter Ethernet
Link-local IPv6 Address . . . . . : fe80::1770:131b:4fc0:17c9%5
IPv4 Address. . . . . : 172.16.9.1
Subnet Mask . . . . . : 255.255.252.0
Default Gateway . . . . . : 172.16.8.1
ipconfig/all
IP Configuration
Host Name . . . . . : DESKTOP-TB99J4B
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : 00-27-0E-13-E0-80
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::1770:131b:4fc0:17c9%5(Preferred)
IPv4 Address. . . . . : 172.16.9.1(Preferred)
Subnet Mask . . . . . : 255.255.252.0
Default Gateway . . . . . : 172.16.8.1
DHCPv6 IAID . . . . . : 301673194
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-7C-8A-13-00-27-0E-13-E0-80
DNS Servers . . . . . : 172.16.8.1
NetBIOS over Tcpip. . . . . : Enabled
ping google.com
Pinging google.com [142.250.66.14] with 32 bytes of data:
Reply from 142.250.66.14: bytes=32 time=4ms TTL=119
Reply from 142.250.66.14: bytes=32 time=4ms TTL=119
Reply from 142.250.66.14: bytes=32 time=5ms TTL=119
Reply from 142.250.66.14: bytes=32 time=5ms TTL=119
Ping statistics for 142.250.66.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),


TCP 172.16.9.1:51133 4.213.25.240:443 ESTABLISHED
TCP 172.16.9.1:51144 142.251.43.238:443 ESTABLISHED
TCP 172.16.9.1:51145 142.250.183.46:443 TIME_WAIT
TCP 0.0.0.0:135 0.0.0.0:0 LISTENING
TCP 0.0.0.0:445 0.0.0.0:0 LISTENING
TCP 0.0.0.0:7000 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49664 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49665 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49666 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49667 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49668 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49669 0.0.0.0:0 LISTENING
TCP 0.0.0.0:49760 0.0.0.0:0 LISTENING
UDP 0.0.0.0:123 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5355 *:*
UDP 0.0.0.0:5353 *:*
UDP 0.0.0.0:5355 *:*
UDP 0.0.0.0:50124 *:*
UDP 0.0.0.0:51726 *:*
UDP 0.0.0.0:51727 *:*
UDP 0.0.0.0:51728 *:*
UDP 0.0.0.0:51729 *:*
UDP 0.0.0.0:51730 *:*
UDP 0.0.0.0:51731 *:*
UDP 0.0.0.0:51732 *:*
UDP 0.0.0.0:51733 *:*
UDP 0.0.0.0:51734 *:*
UDP 0.0.0.0:51735 *:*
UDP 0.0.0.0:51736 *:*
UDP 0.0.0.0:51737 *:*
UDP 0.0.0.0:51738 *:*
UDP

Ex no: 2

The Steps to assigning the ip address manually

AIM

To verify the steps to assigning the ip manually

PROCEDURE:

* Turn on the computer and log in with an administrator account
* In setting, open Network settings
* Identify the active network adapter
* Right click the network adapter and select prop
* In the property select IPV4
* Click the properties button
* Select use the following IP
* Enter the subnet Mask
* Enter the default Gateway
* Select use the following DNS server
* Enter the prefferd DNS server
* Click ok, then click close to save
* open command prompt and run ipconfig to c

the manually assigned IP

INPUT:

computer 1 IP: 192.168.1.1

Subnet mask : 255.255.255.0

Default Gateway: 192.168.1.254

DNS server : 8.8.8.8

OUTPUT:

The IP address were successfully assigned manually to
computers and connectivity b/w the computer was successfully veri

Study of Different types of Network cables and Crimping of cable with RJ45 connector

AIM

To study the different type of network cables and Crimp an ethernet cable using a RJ45 connector

Types of Network Cables

1.) Twisted pair cables
   - → UTP
   - → STP
2.) Coaxial cable
3.) Optical fibre cable

RJ45 color code - TS 68 B

Pin	Color
1	White-orange
2	orange
3	White green
4	Blue
5	white blue
6	Green
7	White brown
8	Brown

Order:

White orange → orange → White green → Blue → White bl
→ Green → White brown → Brown

Ex.5 Design and Configure a simple network using one router and 2 switches

AIM:

To design and configure a simple network using one router, two switches and four PCs in Cisco packet tracer

PROCEDURE:

1. Create the topology with 4 PC, 2 switches (2960) and 1 router
2. connect PC0 and PC1 to Switch 0
3. connect PC2 and PC3 to Switch 1
4. Connect both switches to the router using copper straigh-Through wire
5. Configure the PCs with static IP addresses
6. Configure router interface:
   - → Gi0/0/0 → 192.168.10.1/24
   - → Gi0/0/1 → 192.168.20.1/24
7. Enable both interface using no shutdown command
8. Verify connectivity using ping command

OBSERVATION

⇒ Both LANS were successfully connected through router

⇒ PCs within same LAN communicated successfully

⇒ Successfull ping replies confirmed network connectivity

Ex.no:6 Setting Up a Local Area Network

AIM:

To design and configure a LAN connecting four PCs through a switch, assign static IP addresses, verify connectivity

PROCEDURE:

1. Open Cisco packet tracer
2. place 4 PCs and switch (2960) in the workspace
3. Connect all PCs to the switch using Copper straigh through cables
4. Configure static IP addresses for all PCs
5. Configure the switch management IP through the CLI

Commands:

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

6. Open the command prompt of a PC and use the Ping command to verify connectivity

7. Add a server and configure its IP and HTTP service to demonstrate data sharing

8. Switch to simulation mode and observe the packet flow through the switch

Ex.no:7 Customize Switch with network modules using Cisco packet tracer

AIM:

To customize a modular switch by installing network modules and configure a fiber-optic connection b/w switch and router

PROCEDURE:

1. Open Cisco packet tracer and place - PT-EMPTY, two PCs and a Cisco 2911 router.

2. Open the Switch physical tab and power it OFF

3. Install the required modules
   - → PT-SWITCH-NM-1FFE ⇒ Slot 0 and Slot 1
   - → PT-SWITCH-NM-1FGE ⇒ Slot 2

4. Power ON the switch and Open the CLI

5. Verify the installed interfaces:

   Command:
   ```
   enable
   show ip interface brief
   ```

6. Connect

   PC0 → FastEthernet 0/1 Using copper cable

   PC1 → FastEthernet 1/1 using copper cable

   Switch GigabitEthernet 2/1 → Router using Fiber cable

7. Power OFF the Cisco 2911 router and install the HWIC-

8. Power ON the router and connect the fibre cable to the newly added Gigabit interface.

9. Configure the router interface:

```
enable
configure terminal
interface GigabitEthernet 0/0/0
ip address 192.168.10.1 255.255.0
no shutdown
exit
exit
write memory
```

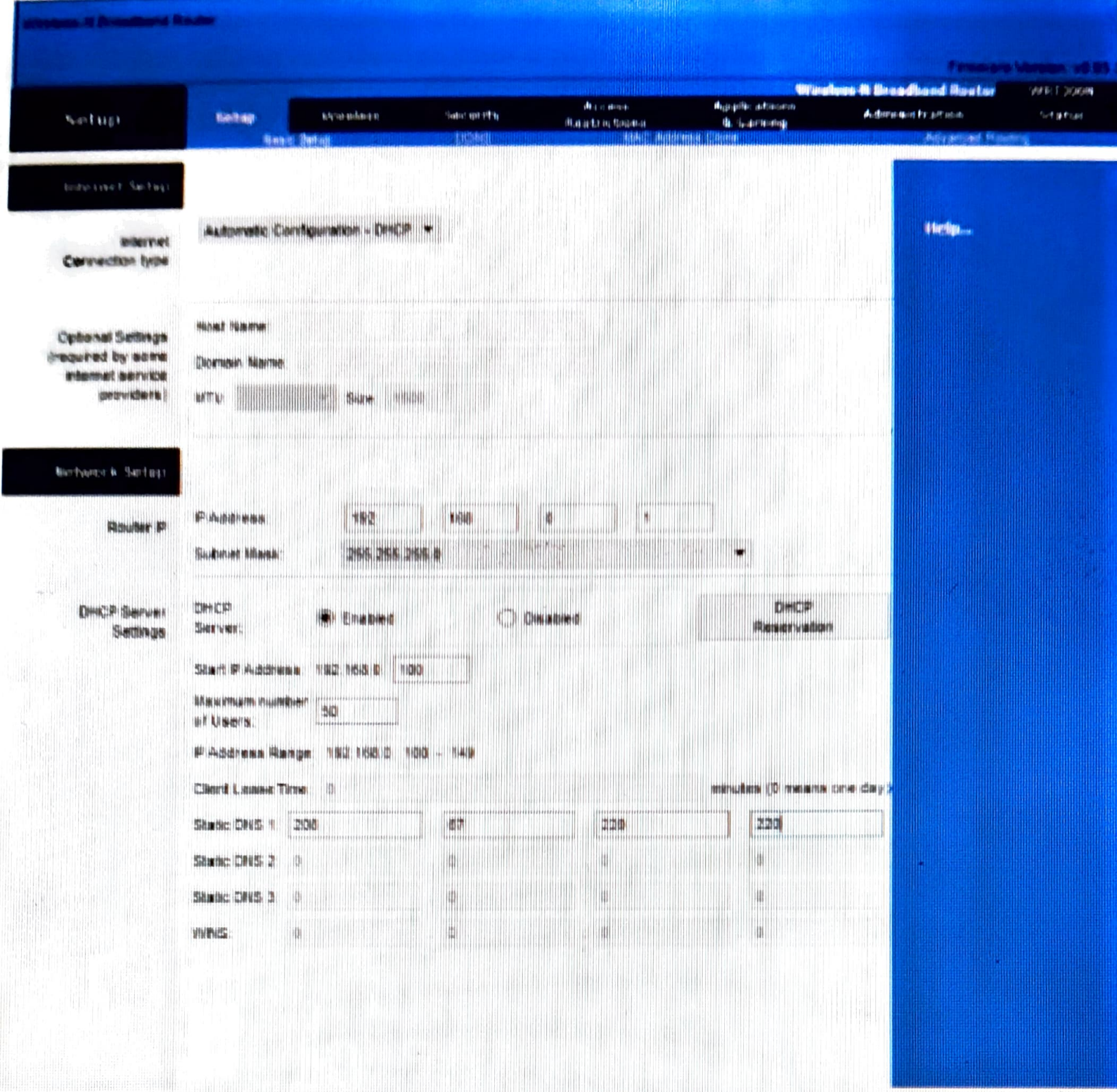
10. Verify the interface:

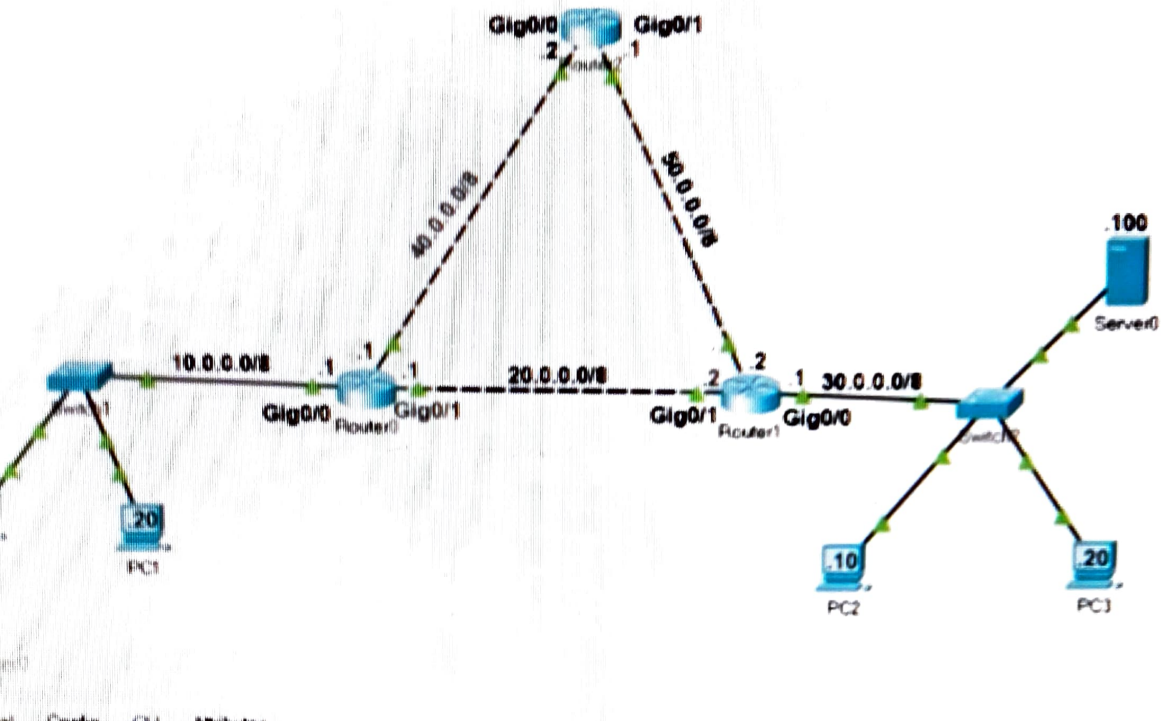
```
show ip interface brief
```

11. Test connectivity using: ping 192.168.10.2

OUTPUT:

The modular switch was successfully customized with copper and fiber interface and a fiber optic connection b/w switch and router was configured successfully


Setup
Internet Setup
Internet Connection type
Automatic Configuration - DHCP
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU:
Size:
Network Setup
Router IP
IP Address:
Subnet Mask:
DHCP Server Settings
DHCP Server:
Enabled
Disabled
DHCP Reservation
Start IP Address:
Maximum number of Users:
IP Address Range:
Client Lease Time:
minutes (0 means one day)
Static DNS 1:
Static DNS 2:
Static DNS 3:
WINS:
Help...
Physical
Config
Desktop
Programming
Attributes
FastEthernet0
Interface



Config CLI Attributes

IOS Command Line Interface

```
ter>enable
ter#configure terminal
er configuration commands, one per line.  End with CNTL/Z.
ter(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2 10
ter(config)#ip route 30.0.0.0 255.0.0.0 40.0.0.2 20
ter(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10
ter(config)#ip route 30.0.0.100 255.255.255.255 20.0.0.2 20
ter(config)#ip route 50.0.0.0 255.0.0.0 40.0.0.2 10
ter(config)#ip route 50.0.0.0 255.0.0.0 20.0.0.2 20
ter(config)#exit
ter#show ip route static
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
       30.0.0.0/8 [10/0] via 20.0.0.2
       30.0.0.100/32 [10/0] via 40.0.0.2
    50.0.0.0/8 [10/0] via 40.0.0.2

ter#
```

Primary route
Backup route
Primary route
Backup route
Primary route
Backup route

Router adds only primary routes to the routing table.

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
Router(config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router(config)#ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router(config)#exit
```